

Advanced (Habanero)

Q1. A bakery sells x cakes and y cookies. They make a profit of

3

per cake and

2

per cookie. If their daily profit is at least

150

, which inequality represents this situation?

- (A) $3x + 2y \geq 150$
- (B) $3x - 2y \geq 150$
- (C) $2x + 3y \geq 150$
- (D) $2x - 3y \geq 150$
- (E) $x + y \geq 150$

Q2. A company produces two products, A and B . The profit per unit of A is

10

and the profit per unit of B is

15

. The company operates under the constraint $2x + 3y \leq 240$, where x and y are the numbers of units of A and B produced, respectively. If the company wants to maximize profit, which of the following is a feasible solution?

- (A) $x = 60, y = 40$
- (B) $x = 40, y = 60$
- (C) $x = 30, y = 50$
- (D) $x = 50, y = 30$
- (E) $x = 20, y = 40$

Q3. A student has at most

20

to spend on pens and pencils. Pens cost

2

each and pencils cost

1

each. If the student buys x pens and y pencils, which inequality represents the cost constraint?

- (A) $2x + y \leq 20$
- (B) $2x - y \leq 20$
- (C) $x + 2y \leq 20$
- (D) $x - 2y \leq 20$
- (E) $x + y \leq 20$

Q4. A factory produces two types of goods, X and Y . The production of X requires at least 3 hours of labor and the production of Y requires at least 2 hours of labor. If the factory has at most 240 labor hours available, which inequality represents this situation?

- (A) $3x + 2y \leq 240$
- (B) $3x - 2y \leq 240$

- (C) $2x + 3y \leq 240$
- (D) $2x - 3y \leq 240$
- (E) $x + y \leq 240$

Q5. A city has a maximum of 5000 tons of waste disposal capacity. The city produces x tons of industrial waste and y tons of residential waste. If industrial waste requires at least 2 times the capacity of residential waste, which inequality represents this situation?

- (A) $x + 2y \leq 5000$
- (B) $2x + y \leq 5000$
- (C) $x - 2y \leq 5000$
- (D) $2x - y \leq 5000$
- (E) $x + y \leq 5000$

Q6. A farmer has at most 100 acres of land to plant x acres of wheat and y acres of barley. Wheat requires at least 2 acres of land per unit and barley requires at least 1 acre of land per unit. Which inequality represents the land constraint?

- (A) $2x + y \leq 100$
- (B) $2x - y \leq 100$
- (C) $x + 2y \leq 100$
- (D) $x - 2y \leq 100$
- (E) $x + y \leq 100$

Q7. A company produces two products, A and B , using two machines, I and II . Machine I can produce at most 40 units of A and 30 units of B per day. Machine II can produce at most 30 units of A and 20 units of B per day. If x and y are the numbers of units of A and B produced, respectively, which system of inequalities represents the production constraints?

- (A) $x \leq 40, y \leq 30$
- (B) $x \leq 30, y \leq 20$
- (C) $x + y \leq 70, 2x + y \leq 90$
- (D) $x + y \leq 60, 2x + y \leq 80$
- (E) $x + y \leq 50, 2x + y \leq 60$

Q8. A student has a maximum of

50

to spend on books and supplies. Books cost

10

each and supplies cost

5

each. If the student buys x books and y supplies, which inequality represents the cost constraint?

- (A) $10x + 5y \leq 50$
- (B) $10x - 5y \leq 50$
- (C) $5x + 10y \leq 50$
- (D) $5x - 10y \leq 50$
- (E) $x + y \leq 50$

Open Ended Questions (FRQ)

Q9. A manufacturer produces two products, X and Y . The profit per unit of X is

20

and the profit per unit of Y is

1

30

. The production of X requires at least 2 hours of labor and the production of Y requires at least 3 hours of labor. The manufacturer has at most 240 labor hours available per day. Find the maximum profit the manufacturer can make per day.

Q10. A shipping company has two types of containers, A and B . Container A has a capacity of 20 cubic meters and container B has a capacity of 30 cubic meters. The company has at most 1000 cubic meters of cargo to ship. The shipping cost per container A is

100

and the shipping cost per container B is

150

. Find the maximum number of containers the company can ship within the given cargo capacity and cost constraints.

Chef's Tips

- Tip 1: Ensure students understand the context of the problem before applying mathematical concepts.
- Tip 2: Encourage students to visualize the feasible region for systems of inequalities.
- Tip 3: Remind students to consider the constraints and the objective function when optimizing within a feasible region.